Connected Histories

Computing, the Internet and Artificial Intelligence, from Babbage to Transformers

A reference reader on the people, institutions, technologies and events behind modern computing and AI

About this document. This reader summarizes well-known history for testing retrieval-augmented generation (RAG) and knowledge-graph (GraphRAG) systems. Its sections are written so that people, institutions and technologies recur across chapters: many questions can only be answered by connecting facts from different sections. It is a condensed overview, so dates and attributions are simplified; consult primary sources before citing it.

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Overview

Modern artificial intelligence sits on top of nearly two centuries of connected work. The mathematics of logic and graphs, the theory of what machines can compute, the electronic computers built during the Second World War, the transistor and the integrated circuit, the networks that became the internet, and successive waves of AI research all feed into today's language models and retrieval systems. The same institutions appear again and again: Bell Labs, the National Physical Laboratory, Princeton, MIT, Stanford, IBM, the University of Toronto and Google. This reader follows those threads.

1. Foundations: Graphs, Logic and Mechanical Computation

Euler and the Seven Bridges of Königsberg

In 1736 the Swiss mathematician Leonhard Euler studied a puzzle from the Prussian city of Königsberg: could a walk cross each of the city's seven bridges exactly once? Euler showed that it could not by reducing the map to land masses (vertices) connected by bridges (edges). His solution is regarded as the founding result of graph theory, the mathematics that underlies graph databases and knowledge graphs today (see Section 9).

Boole and the algebra of logic

The English mathematician George Boole published *The Mathematical Analysis of Logic* in 1847 and *An Investigation of the Laws of Thought* in 1854. Boolean algebra treats logical statements as values that are either true or false and combines them with AND, OR and NOT. Nearly a century later Claude Shannon showed that Boolean algebra could describe electrical switching circuits (Section 2).

Jacquard, Babbage and Lovelace

Joseph Marie Jacquard's loom of 1804 used chains of punched cards to control the pattern woven into fabric. Charles Babbage, an English mathematician, began the Difference Engine in the 1820s to calculate mathematical tables, and in 1837 described the far more ambitious Analytical Engine. The Analytical Engine separated a 'mill' that performed arithmetic from a 'store' that held numbers, which is the same division between processor and memory used in modern computers, and it borrowed Jacquard's punched cards for its programs. It was never completed.

Ada Lovelace, the daughter of the poet Lord Byron, translated an 1842 article on the Analytical Engine by the Italian engineer Luigi Menabrea and published it in 1843 with her own extensive notes. Note G described a step-by-step method for the Engine to compute Bernoulli numbers and is often called the first published computer program. Lovelace also suggested that such a machine might manipulate symbols other than numbers, such as music.

Hollerith and the origins of IBM

Herman Hollerith applied punched cards to data processing. His electromechanical tabulating machines processed the 1890 United States Census far faster than hand counting. Hollerith's Tabulating Machine Company merged with other firms in 1911 to form the Computing-Tabulating-Recording Company (CTR), which was renamed International Business Machines (IBM) in 1924 under Thomas J. Watson. IBM reappears throughout this reader: at the Dartmouth workshop, in the invention of the relational database and in the chess match against Garry Kasparov.

2. The Theory of Computation

Hilbert's decision problem

In 1928 the German mathematician David Hilbert posed the *Entscheidungsproblem* (decision problem): is there a mechanical procedure that can decide whether any given mathematical statement is provable? Kurt Gödel's incompleteness theorems of 1931 had already shown deep limits to formal systems, and in 1936 two mathematicians independently showed that the answer to Hilbert's question was no.

Church and Turing

Alonzo Church of Princeton University answered the question using his lambda calculus. Alan Turing, a young fellow of King's College, Cambridge, answered it in his 1936 paper *On Computable Numbers, with an Application to the Entscheidungsproblem*. Turing imagined a simple machine that reads and writes symbols on an endless tape, and showed that a single 'universal' machine could imitate any other. The

Turing machine became the standard model of computation. Turing then went to Princeton, where he completed his PhD under Church in 1938, before returning to England shortly before the Second World War.

Shannon: switching circuits

In 1937 Claude Shannon, a master's student at the Massachusetts Institute of Technology (MIT), wrote *A Symbolic Analysis of Relay and Switching Circuits*. It showed that Boole's algebra could be used to design and simplify circuits of relays and switches, which made digital circuit design a mathematical discipline. Shannon joined Bell Telephone Laboratories, where he later founded information theory (Section 4). In 1943 Turing visited Bell Labs to work on secure speech systems, and he and Shannon met and discussed the idea of thinking machines.

3. Computers at War and the First Electronic Machines

Bletchley Park and the Bombe

During the Second World War Britain's codebreakers worked at Bletchley Park. Building on the work of Polish cryptologists, including Marian Rejewski's 'bomba' machine, Alan Turing designed the British Bombe, an electromechanical device for finding the daily settings of the German Enigma cipher machine. The first Bombe went into service in 1940, and Gordon Welchman's 'diagonal board' made it much more effective.

Colossus

German high command used a different, more complex cipher machine, the Lorenz, codenamed 'Tunny' by the British. To attack it, Tommy Flowers, an engineer at the Post Office Research Station at Dollis Hill, designed Colossus, which used about 1,600 vacuum tubes in its first version and became operational at Bletchley Park in early 1944. The section that ran it was led by the mathematician Max Newman. Colossus is regarded as the first programmable electronic digital computer, but it was kept secret for decades after the war.

ENIAC and the stored-program idea

In the United States, J. Presper Eckert and John Mauchly built the Electronic Numerical Integrator and Computer (ENIAC) at the University of Pennsylvania's Moore School of Electrical Engineering. Funded by the US Army to calculate artillery firing tables, ENIAC contained about 18,000 vacuum tubes and was unveiled in February 1946. Its programmers, among them Jean Jennings Bartik and Betty Snyder Holberton, configured it by setting switches and plugging cables.

John von Neumann of the Institute for Advanced Study in Princeton consulted on ENIAC's successor, EDVAC. His 1945 *First Draft of a Report on the EDVAC* described a computer that stores its program in the same memory as its data. This 'von Neumann architecture' echoes the mill-and-store design of Babbage's Analytical Engine and is still the basic organization of most computers.

The first stored-program computers

The first program stored in electronic memory ran on 21 June 1948 on the Manchester Baby, built at the University of Manchester by Frederic Williams, Tom Kilburn and Geoff Tootill. In Britain, Turing had already designed a stored-program computer, the Automatic Computing Engine (ACE), at the National Physical Laboratory (NPL) in 1945–46; he moved to Manchester in 1948. In 1950 he published *Computing Machinery and Intelligence*, which proposed the 'imitation game', now called the Turing test. Eckert and Mauchly went on to build UNIVAC I, the first commercial computer produced in the United States, delivered to the US Census Bureau in 1951, about sixty years after Hollerith's machines had transformed the census.

4. Bell Labs and the Silicon Revolution

The transistor

Vacuum tubes were large, hot and unreliable. In December 1947 John Bardeen and Walter Brattain at Bell Labs demonstrated the point-contact transistor, a solid-state amplifier and switch. Their group leader, William Shockley, developed the junction transistor soon after. The three shared the 1956 Nobel Prize in Physics.

Information theory

In 1948, the same year the transistor was announced, Claude Shannon published *A Mathematical Theory of Communication* at Bell Labs. It defined information mathematically, set limits on how much data a noisy channel can carry, and popularized the word 'bit' (binary digit), a term Shannon credited to his colleague John Tukey. Information theory underpins data compression, error correction and digital communication.

From Shockley to Fairchild to Intel

Shockley left Bell Labs and in 1956 founded Shockley Semiconductor Laboratory in Mountain View, California, the beginning of what became Silicon Valley. His management style drove eight young researchers, later nicknamed the 'traitorous eight', to leave in 1957 and found Fairchild Semiconductor. Among them were Robert Noyce and Gordon Moore.

The integrated circuit, which places many transistors on a single chip, was invented twice. Jack Kilby demonstrated the first integrated circuit at Texas Instruments in 1958, and Robert Noyce at Fairchild designed a practical monolithic silicon version in 1959 using the planar process developed by his colleague Jean Hoerni. Kilby received the Nobel Prize in Physics in 2000; Noyce had died in 1990.

In 1965, while at Fairchild, Gordon Moore observed in the magazine *Electronics* that the number of components on a chip was doubling at a regular pace; he later revised the rate to about every two years. This 'Moore's law' guided the industry for half a century. In 1968 Noyce and Moore left Fairchild to found Intel, joined by Andrew Grove. Intel's 4004 of 1971, designed by Federico Faggin, Ted Hoff, Stanley Mazor and Masatoshi Shima for the Japanese calculator maker Busicom, was the first commercial microprocessor.

Unix and C

Bell Labs also shaped software. Ken Thompson and Dennis Ritchie created the Unix operating system in 1969, and Ritchie developed the C programming language in the early 1970s. Both shared the 1983 Turing Award. Bell Labs returns later in this story as the place where Yann LeCun built convolutional neural networks (Section 7).

5. Networking: From ARPANET to the World Wide Web

ARPA and Licklider

After the Soviet Union launched Sputnik in 1957, the United States created the Advanced Research Projects Agency (ARPA) in 1958. In 1962 J. C. R. Licklider, a psychologist who had worked at MIT and at the consulting firm Bolt Beranek and Newman (BBN), became the first head of ARPA's Information Processing Techniques Office. Licklider promoted interactive computing and imagined an 'intergalactic computer network' linking researchers.

Packet switching

Two researchers independently invented packet switching, which splits messages into small blocks that travel separately across a network. Paul Baran at the RAND Corporation proposed it in the early 1960s

as a way to build communications that could survive damage, and Donald Davies at Britain's National Physical Laboratory, the same laboratory where Turing had designed the ACE, developed it in the mid-1960s and coined the word 'packet'.

ARPANET

ARPANET used packet switching to connect research computers. BBN built its packet switches, called Interface Message Processors. The first message was sent on 29 October 1969 from Leonard Kleinrock's laboratory at the University of California, Los Angeles (UCLA) to the Stanford Research Institute (SRI), where the second node was hosted by Douglas Engelbart's Augmentation Research Center. The system crashed after the first two letters, 'LO', of the word 'LOGIN'. Engelbart had become famous the year before for his December 1968 demonstration, later called the 'Mother of All Demos', which introduced the computer mouse, hypertext links and video conferencing.

TCP/IP

As more networks appeared, they needed a common language. Vint Cerf and Bob Kahn published the design of the Transmission Control Protocol in 1974, which evolved into the TCP/IP protocol suite. ARPANET switched to TCP/IP on 1 January 1983, a date often treated as the birth of the internet. Cerf and Kahn received the 2004 Turing Award.

The World Wide Web

In March 1989 Tim Berners-Lee, a British scientist at CERN, the European particle physics laboratory near Geneva, proposed a hypertext system for sharing information among physicists. He wrote the first web server and browser, his colleague Robert Cailliau helped develop and promote the project, and the Web became publicly available in 1991. CERN released the Web software into the public domain in 1993. The Web combined Engelbart-style hypertext with the internet's TCP/IP. Berners-Lee received the 2016 Turing Award.

6. The Birth of Artificial Intelligence and the AI Winters

The Dartmouth workshop

In 1955 John McCarthy, then at Dartmouth College, wrote a proposal for a summer research project together with Marvin Minsky of Harvard, Nathaniel Rochester of IBM and Claude Shannon of Bell Labs. The proposal introduced the term 'artificial intelligence', and the Dartmouth Summer Research Project on Artificial Intelligence took place in 1956. It is generally considered the founding event of AI as a field.

At Dartmouth, Allen Newell and Herbert Simon presented the Logic Theorist, written with Cliff Shaw of the RAND Corporation, which proved theorems from *Principia Mathematica* and is often called the first AI program. Newell and Simon shared the 1975 Turing Award.

Lisp and the first AI laboratories

McCarthy invented the Lisp programming language at MIT in 1958; it became the main language of AI research for decades. McCarthy and Minsky founded the MIT Artificial Intelligence Project in 1959. McCarthy moved to Stanford University in 1962 and founded the Stanford Artificial Intelligence Laboratory (SAIL). Minsky received the Turing Award in 1969 and McCarthy in 1971.

The perceptron and its critics

A different approach tried to imitate the brain. In 1958 Frank Rosenblatt of the Cornell Aeronautical Laboratory introduced the perceptron, a simple artificial neural network that learned from examples, and built it in hardware as the Mark I Perceptron. In 1969 Marvin Minsky and Seymour Papert published *Perceptrons*, which proved that single-layer perceptrons cannot learn some simple functions, such as exclusive-or. The book contributed to a long decline in neural network research.

AI winters

Early optimism outran results. In the United Kingdom, the 1973 Lighthill Report criticized AI research and led to deep funding cuts. Periods of reduced funding and interest in the 1970s and again in the late 1980s, after the collapse of the market for expert systems and Lisp machines, became known as 'AI winters'.

Deep Blue

Symbolic search-based AI achieved a famous milestone in May 1997, when IBM's Deep Blue chess computer defeated world champion Garry Kasparov in a six-game match. Deep Blue relied on specialized hardware and massive search rather than learning, a contrast with the neural network methods described next.

7. The Return of Neural Networks

Hopfield networks and backpropagation

In 1982 the physicist John Hopfield described a recurrent neural network that stores memories as stable patterns, now called a Hopfield network. In 1986 David Rumelhart, Geoffrey Hinton and Ronald Williams published a paper in *Nature* showing how the backpropagation algorithm lets multi-layer networks learn useful internal representations. Multi-layer networks trained with backpropagation overcame the limits that Minsky and Papert had identified in single-layer perceptrons. Hinton, a British-Canadian researcher, later built his group at the University of Toronto.

Convolutional networks at Bell Labs

Yann LeCun, a French computer scientist, worked as a postdoctoral researcher with Hinton in Toronto before joining Bell Labs in 1988. There he developed convolutional neural networks trained with backpropagation, which read handwritten digits; versions of this system, known as LeNet, were used to read zip codes and bank cheques. LeCun later became a professor at New York University and in 2013 the founding director of Facebook AI Research.

Recurrent networks and LSTM

To process sequences such as text and speech, Sepp Hochreiter and Jürgen Schmidhuber introduced the long short-term memory (LSTM) network in 1997. LSTMs remained the leading sequence model until the Transformer (Section 8). In Montreal, Yoshua Bengio's group at the Université de Montréal, later the Mila institute, worked on neural language models and sequence learning.

GPUs, ImageNet and AlexNet

Two developments made deep learning practical. First, NVIDIA, founded in 1993 by Jensen Huang, Chris Malachowsky and Curtis Priem to make graphics chips, released the CUDA platform in 2007, which let researchers program graphics processing units (GPUs) for general computation. Second, Fei-Fei Li began building ImageNet at Princeton and continued it at Stanford; released in 2009, it eventually contained more than 14 million labelled images and became the basis of an annual competition, the ImageNet Large Scale Visual Recognition Challenge.

In 2012 Alex Krizhevsky, Ilya Sutskever and Geoffrey Hinton of the University of Toronto entered AlexNet, a deep convolutional network trained on two NVIDIA GPUs. It achieved a top-5 error rate of 15.3%, against 26.2% for the second-best entry. The result convinced much of the field to adopt deep learning. In 2013 Google acquired DNNresearch, the company Hinton had formed with Krizhevsky and Sutskever.

Google Brain, word2vec and DeepMind

Google Brain was started in 2011 by Jeff Dean, Greg Corrado and Andrew Ng of Stanford University. In 2013 Tomas Mikolov and colleagues at Google released word2vec, which represented words as vectors so that words with similar meanings lie close together, an early form of the embeddings used in vector search today (Section 9).

DeepMind was founded in London in 2010 by Demis Hassabis, Shane Legg and Mustafa Suleyman and acquired by Google in 2014. Its AlphaGo program defeated the Go champion Lee Sedol 4 games to 1 in Seoul in March 2016, a result many experts had expected to be a decade away. DeepMind's AlphaFold later predicted protein structures with high accuracy, and Hassabis and John Jumper shared half of the 2024 Nobel Prize in Chemistry for it; the other half went to David Baker. In 2023 Google merged Google Brain and DeepMind into a single unit, Google DeepMind.

8. Attention, Transformers and Large Language Models

Sequence-to-sequence learning and attention

In 2014 Ilya Sutskever, Oriol Vinyals and Quoc Le at Google showed that LSTM networks could translate between languages by encoding a whole sentence into a vector and decoding it. The same year, Dzmitry Bahdanau, Kyunghyun Cho and Yoshua Bengio in Montreal introduced an attention mechanism that let the decoder look back at the most relevant words of the source sentence instead of relying on a single vector.

The Transformer

In 2017 eight researchers, Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan Gomez, Lukasz Kaiser and Illia Polosukhin, published *Attention Is All You Need* at the NIPS conference (now NeurIPS). Most worked at Google Brain or Google Research; Gomez was a University of Toronto student working at Google Brain. The paper's Transformer architecture dropped recurrence entirely and relied on self-attention, which allowed training to run in parallel on GPUs and other accelerators. It reported state-of-the-art results on English-to-German and English-to-French translation. Several authors later founded companies; for example, Gomez co-founded Cohere and Shazeer co-founded Character.AI.

BERT, GPT and OpenAI

The Transformer became the basis of nearly all large language models. OpenAI, founded in December 2015 by a group including Sam Altman, Elon Musk, Greg Brockman and Ilya Sutskever, who became its chief scientist, released the first Generative Pre-trained Transformer (GPT) in 2018. Later that year Google released BERT, a Transformer encoder pre-trained on large text collections that greatly improved search and language understanding. OpenAI's GPT-3 of 2020 had 175 billion parameters, and ChatGPT, released on 30 November 2022, brought large language models to a mass audience.

The thread from AlexNet to ChatGPT runs through a single person: Ilya Sutskever co-authored AlexNet in Hinton's Toronto lab, co-authored the 2014 sequence-to-sequence paper at Google, and co-founded OpenAI.

9. Data, Graphs and Retrieval: From Relational Databases to GraphRAG

The relational model

In 1970 Edgar F. Codd, a British computer scientist at IBM's San Jose research laboratory, published *A Relational Model of Data for Large Shared Data Banks*. It organized data into tables related by shared values. IBM researchers Donald Chamberlin and Raymond Boyce then designed SEQUEL, later renamed SQL, to query relational databases. Codd received the 1981 Turing Award.

Graph databases and Neo4j

Relational databases answer questions about connected data through joins, which become slow and complex when relationships are many levels deep. Graph databases store relationships directly, applying Euler's vertices and edges (Section 1) as nodes and relationships. Neo4j, developed by a Swedish team including Emil Eifrem, became the most widely used graph database. Its query language, Cypher, introduced in 2011, describes graph patterns using ASCII-art arrows such as (a)-[:KNOWS]->(b), and it strongly influenced GQL, the graph query language published as an ISO standard in 2024. In 2023 Neo4j added vector indexes, allowing one database to support both graph traversal and similarity search.

Knowledge graphs

In 2012 Google introduced its Knowledge Graph, which links people, places and things so that search results can show facts rather than only links. It drew partly on Freebase, a collaborative database built by Metaweb, which Google had acquired in 2010. Knowledge graphs store facts as entities connected by typed relationships, such as *Turing* STUDIED_UNDER *Church*.

Sentence embeddings

Embeddings turn text into vectors so that similar meanings produce nearby vectors, extending the idea behind word2vec from words to whole sentences. In 2019 Nils Reimers and Iryna Gurevych of the Technical University of Darmstadt introduced Sentence-BERT, which adapted BERT to produce sentence embeddings efficiently, and released the sentence-transformers library. A popular model from that library, all-MiniLM-L6-v2, is based on Microsoft's MiniLM architecture, has six Transformer layers and maps text to 384-dimensional vectors.

Retrieval-augmented generation

Language models can produce fluent but unsupported answers. In 2020 Patrick Lewis and colleagues at Facebook AI Research, University College London and New York University introduced retrieval-augmented generation (RAG), in which a system first retrieves relevant passages and then generates an answer conditioned on them.

GraphRAG

Vector search retrieves passages that look similar to the question, but it struggles with questions whose answer is spread across several documents or requires following a chain of relationships. In 2024 researchers at Microsoft Research, led by Darren Edge, described GraphRAG, which uses a language model to extract entities and relationships from documents into a knowledge graph and uses that graph, together with summaries of its communities, to answer broader questions. GraphRAG thus brings together several threads of this reader: Euler's graphs, knowledge graphs, Transformer-based language models and embedding-based retrieval.

10. Recognition: The Turing Award and the Nobel Prizes

The Association for Computing Machinery (ACM) established the A.M. Turing Award in 1966, named after Alan Turing; it is often called the Nobel Prize of computing. Its first recipient was Alan Perlis. Many people in this reader have received it.

Award year	Recipient(s)	Recognized for (summary)
1969	Marvin Minsky	Artificial intelligence
1971	John McCarthy	Artificial intelligence; Lisp
1975	Allen Newell, Herbert Simon	AI, cognitive psychology and list processing
1981	Edgar F. Codd	Relational databases

Award year	Recipient(s)	Recognized for (summary)
1983	Ken Thompson, Dennis Ritchie	Unix and operating systems theory
2004	Vint Cerf, Bob Kahn	TCP/IP and internet architecture
2016	Tim Berners-Lee	The World Wide Web
2018	Geoffrey Hinton, Yann LeCun, Yoshua Bengio	Deep neural networks

Computing has also been recognized by Nobel Prizes in the sciences:

Year	Prize	Recipient(s)	Work
1956	Physics	John Bardeen, Walter Brattain, William Shockley	The transistor (Bell Labs)
2000	Physics	Jack Kilby (shared)	The integrated circuit (Texas Instruments)
2024	Physics	John Hopfield, Geoffrey Hinton	Foundations of machine learning with artificial neural networks
2024	Chemistry	Demis Hassabis, John Jumper (shared with David Baker)	Protein structure prediction (AlphaFold)

Geoffrey Hinton is therefore one of very few people to have received both the Turing Award and a Nobel Prize.

11. Recurring Institutions

Several institutions connect otherwise separate chapters of this history:

- **Bell Labs** – the transistor (Bardeen, Brattain, Shockley), information theory (Shannon), Unix and C (Thompson, Ritchie), Turing's 1943 visit, and LeCun's convolutional networks.
- **National Physical Laboratory (UK)** – Turing's ACE design and Donald Davies' packet switching.
- **Princeton** – Church's lambda calculus, Turing's PhD, von Neumann at the Institute for Advanced Study, and the beginnings of ImageNet under Fei-Fei Li.
- **MIT** – Shannon's 1937 thesis, Lisp, the AI Project of McCarthy and Minsky, and Licklider's early work.
- **Stanford** – McCarthy's SAIL, Fei-Fei Li's ImageNet, Andrew Ng; the Stanford Research Institute (SRI) hosted Engelbart and ARPANET's second node.
- **IBM** – descended from Hollerith's company; Rochester at Dartmouth; Codd's relational model; Deep Blue.
- **RAND Corporation** – Baran's packet switching and Shaw's work on the Logic Theorist.
- **University of Toronto** – Hinton's group, LeCun's postdoc, AlexNet, Sutskever, and Transformer co-author Aidan Gomez.
- **Google** – DNNresearch, Google Brain, word2vec, DeepMind, sequence-to-sequence learning, the Transformer, BERT and the Knowledge Graph.

12. Reference Tables

12.1 People

Person	Main affiliation(s)	Key contributions	Connections
Leonhard Euler	Academies of St Petersburg and Berlin	Graph theory (1736)	Foundation of graph databases
George Boole	Queen's College, Cork	Boolean algebra (1847, 1854)	Applied to circuits by Shannon
Charles Babbage	Cambridge	Difference and Analytical Engines	Worked with Lovelace; used Jacquard cards
Ada Lovelace	–	Notes on the Analytical Engine (1843)	Translated Menabrea; daughter of Byron
Herman Hollerith	Tabulating Machine Company	Punched-card tabulator, 1890 census	Company became IBM
Alonzo Church	Princeton	Lambda calculus (1936)	Turing's PhD supervisor
Alan Turing	Cambridge, Bletchley Park, NPL, Manchester	Turing machine, Bombe, ACE, Turing test	Studied under Church; met Shannon; Turing Award named after him
Claude Shannon	MIT, Bell Labs	Switching circuits (1937), information theory (1948)	Co-proposed Dartmouth workshop; met Turing
John von Neumann	Institute for Advanced Study	Stored-program architecture (1945)	Consulted on ENIAC/EDVAC
Tommy Flowers	Post Office Research Station	Colossus (1944)	Worked for Bletchley Park
Eckert and Mauchly	Moore School, University of Pennsylvania	ENIAC (1946), UNIVAC I (1951)	UNIVAC delivered to Census Bureau
William Shockley	Bell Labs, Shockley Semiconductor	Junction transistor	His staff founded Fairchild
Robert Noyce	Fairchild, Intel	Monolithic integrated circuit	Co-founded Intel with Moore
Gordon Moore	Fairchild, Intel	Moore's law (1965)	One of the 'traitorous eight'
Jack Kilby	Texas Instruments	First integrated circuit (1958)	Nobel Prize 2000
Ken Thompson, Dennis Ritchie	Bell Labs	Unix, C	Turing Award 1983
J. C. R. Licklider	MIT, BBN, ARPA	Vision of networked computing	First head of ARPA's IPTO
Donald Davies	National Physical Laboratory	Packet switching; coined 'packet'	Same lab as Turing's ACE
Paul Baran	RAND Corporation	Packet switching	Independent of Davies
Leonard Kleinrock	UCLA	First ARPANET message (1969)	Sent to SRI
Douglas Engelbart	SRI	Mouse, hypertext, 1968 demo	Hosted ARPANET's second node
Vint Cerf, Bob Kahn	Stanford, ARPA	TCP/IP (1974)	Turing Award 2004
Tim Berners-Lee	CERN	World Wide Web (1989–91)	Worked with Robert Cailliau; Turing Award 2016
John McCarthy	Dartmouth, MIT, Stanford	Coined 'artificial intelligence'; Lisp	Founded SAIL

Person	Main affiliation(s)	Key contributions	Connections
Marvin Minsky	Harvard, MIT	Co-founded MIT AI Project; *Perceptrons*	Dartmouth co-proposer
Newell and Simon	RAND, Carnegie Tech	Logic Theorist (1956)	Turing Award 1975
Frank Rosenblatt	Cornell Aeronautical Laboratory	Perceptron (1958)	Criticized by Minsky and Papert
Edgar F. Codd	IBM	Relational model (1970)	Turing Award 1981
John Hopfield	Caltech, Princeton	Hopfield network (1982)	Nobel Prize 2024 with Hinton
Geoffrey Hinton	University of Toronto, Google	Backpropagation (1986), AlexNet (2012)	Supervised LeCun (postdoc) and Sutskever; Turing Award 2018; Nobel 2024
Yann LeCun	Bell Labs, NYU, Facebook AI Research	Convolutional networks, LeNet	Turing Award 2018
Yoshua Bengio	Université de Montréal, Mila	Neural language models, attention (2014)	Turing Award 2018
Fei-Fei Li	Princeton, Stanford	ImageNet (2009)	Dataset used by AlexNet
Ilya Sutskever	University of Toronto, Google, OpenAI	AlexNet, sequence-to-sequence learning	Co-founded OpenAI
Demis Hassabis	DeepMind	AlphaGo, AlphaFold	Nobel Prize in Chemistry 2024
Ashish Vaswani et al.	Google Brain, Google Research	The Transformer (2017)	Built on attention from Bengio's group
Nils Reimers, Iryna Gurevych	TU Darmstadt	Sentence-BERT (2019)	sentence-transformers library
Patrick Lewis et al.	Facebook AI Research, UCL, NYU	Retrieval-augmented generation (2020)	
Darren Edge et al.	Microsoft Research	GraphRAG (2024)	Combines knowledge graphs and LLMs

12.2 Technologies and their lineage

Technology	Year	Created by	Built on or led to
Graph theory	1736	Leonhard Euler	Knowledge graphs, Neo4j, GraphRAG
Boolean algebra	1847–54	George Boole	Shannon's switching circuits
Analytical Engine	1837	Charles Babbage	Jacquard punched cards; anticipated von Neumann architecture
Turing machine	1936	Alan Turing	Answered Hilbert's decision problem
Bombe	1940	Alan Turing, Gordon Welchman	Built on Rejewski's Polish bomba
Colossus	1944	Tommy Flowers	Attacked the Lorenz cipher
ENIAC	1946	Eckert, Mauchly	Led to EDVAC and UNIVAC I
Transistor	1947	Bardeen, Brattain, Shockley	Integrated circuit
Integrated circuit	1958–59	Kilby, Noyce	Microprocessor; Moore's law
Intel 4004	1971	Faggin, Hoff, Mazor, Shima	First commercial microprocessor

Technology	Year	Created by	Built on or led to
ARPANET	1969	ARPA, BBN, UCLA, SRI	Packet switching; led to the internet
TCP/IP	1974–83	Cerf, Kahn	The internet; the Web
World Wide Web	1989–91	Tim Berners-Lee	Hypertext over TCP/IP
Lisp	1958	John McCarthy	Main AI language for decades
Perceptron	1958	Frank Rosenblatt	Multi-layer networks and backpropagation
Relational model / SQL	1970–74	Codd; Chamberlin and Boyce	Contrast with graph databases
Backpropagation	1986	Rumelhart, Hinton, Williams	Deep learning
LSTM	1997	Hochreiter, Schmidhuber	Sequence-to-sequence models; replaced by Transformers
CUDA	2007	NVIDIA	GPU training of AlexNet
ImageNet	2009	Fei-Fei Li	AlexNet breakthrough (2012)
word2vec	2013	Tomas Mikolov et al. (Google)	Sentence embeddings
Attention mechanism	2014	Bahdanau, Cho, Bengio	The Transformer
Transformer	2017	Vaswani et al. (Google)	BERT, GPT, ChatGPT, Sentence-BERT
Cypher	2011	Neo4j	Influenced the ISO GQL standard (2024)
Sentence-BERT	2019	Reimers, Gurevych	all-MiniLM-L6-v2 and vector search
RAG	2020	Lewis et al.	GraphRAG
GraphRAG	2024	Microsoft Research	Knowledge graphs + language models

12.3 Timeline

Year	Event
1736	Euler solves the Seven Bridges of Königsberg problem
1804	Jacquard's punched-card loom
1837	Babbage describes the Analytical Engine
1843	Lovelace publishes her notes, including Note G
1854	Boole's *Laws of Thought*
1890	Hollerith's tabulators process the US Census
1924	CTR is renamed IBM
1936	Turing's *On Computable Numbers*; Church's lambda calculus
1937	Shannon's master's thesis on switching circuits
1940	First British Bombe in service at Bletchley Park
1944	Colossus operational at Bletchley Park
1945	Von Neumann's *First Draft of a Report on the EDVAC*
1946	ENIAC unveiled
1947	Transistor demonstrated at Bell Labs
1948	Shannon's information theory; Manchester Baby runs the first stored program

Year	Event
1950	Turing proposes the imitation game
1956	Dartmouth workshop; Logic Theorist; Nobel Prize for the transistor
1957	'Traitorous eight' found Fairchild Semiconductor; Sputnik launched
1958	ARPA founded; Kilby's integrated circuit; Lisp; Rosenblatt's perceptron
1965	Moore's law
1966	ACM establishes the Turing Award
1968	Intel founded; Engelbart's 'Mother of All Demos'
1969	First ARPANET message; Unix; *Perceptrons* published
1970	Codd's relational model
1971	Intel 4004 microprocessor
1973	Lighthill Report
1974	Cerf and Kahn publish TCP
1983	ARPANET switches to TCP/IP
1986	Backpropagation paper in *Nature*
1989	Berners-Lee proposes the Web; LeCun's zip-code reader at Bell Labs
1997	Deep Blue defeats Kasparov; LSTM introduced
2007	NVIDIA CUDA released
2009	ImageNet released
2011	Google Brain founded; Cypher introduced
2012	AlexNet wins ImageNet challenge; Google Knowledge Graph
2013	Google acquires DNNresearch; word2vec
2014	Google acquires DeepMind; sequence-to-sequence and attention papers
2015	OpenAI founded
2016	AlphaGo defeats Lee Sedol
2017	*Attention Is All You Need* introduces the Transformer
2018	GPT and BERT; Turing Award to Hinton, LeCun and Bengio
2019	Sentence-BERT
2020	GPT-3; retrieval-augmented generation
2022	ChatGPT released
2023	Google DeepMind formed; Neo4j adds vector indexes
2024	GraphRAG; ISO GQL standard; Nobel Prizes to Hopfield, Hinton, Hassabis and Jumper